Comparison of exponential functions

1 The exponential function J is given by $y = 9\left(3^{-x}\right)$.
The following table shows the function values for exponential function K:

x	-1	1	2	5	10
y	15	$\frac{5}{3}$	$\frac{5}{9}$	$\frac{5}{243}$	$\frac{5}{59049}$

a Determine if the exponential function K is increasing or decreasing.

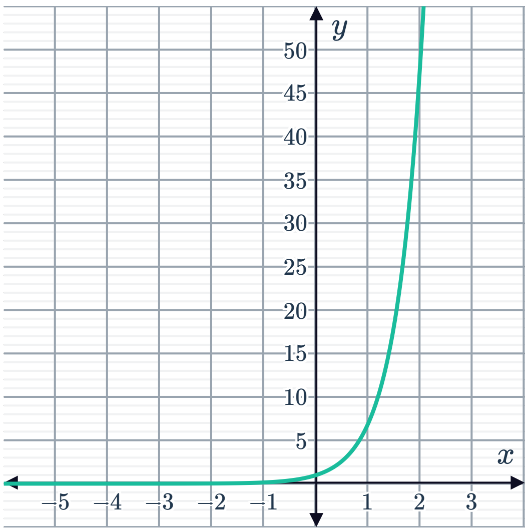
b Graph the exponential functions J and K on the same coordinate plane.

c Describe the end behavior of each function as $x \to \infty$.

d Determine what transformation we can apply to the function $y = 3^{-x}$ to produce function K.

2 Consider the following pair of functions:

- Function P



- Function Q
  $y = 2^x + 4$

a Determine how many times the graphs of P and Q intersect.

b Determine which function tends to infinity the quickest as $x \to \infty$.

3 The exponential function E is given by $y = 3^x + 4$ and the exponential function F is given by $y = -8^x + 1$.

a Graph the exponential functions on the same coordinate plane.

b Identify how many times E and F intersect.

c Describe the end behavior of each exponential function as $x \to \infty$.

Applications

4 Switzerland's population in the next 10 years is expected to grow approximately according to the model $P = 8\,(1 + r)^t$, where P represents the population (in millions) t years from now.
The world population in the next 10 years is expected to grow approximately according to the model $Q = 7130\,(1.0133)^t$, where Q represents the world population (in millions) t years from now.

- **a** Use the model for P to find the current population in Switzerland.
- **b** Use the model for Q to find:
  - **i** The current world population.
  - **ii** The population of the world in twenty years time, to the nearest million.

5 Bank A pays interest at a rate of 5.3% p.a. compounded annually, while Bank B offers a rate of 5% p.a. with interest compounded quarterly.

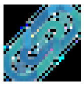
- **a** Form an equation for A, the amount $\$P$ would grow to after t years if invested with:
  - **i** Bank A.
  - **ii** Bank B.
- **b** State the effective annual rate for Bank B as a percentage, to two decimal places.
- **c** Which bank offers the better deal?

6 The median national sale price of property over the next 10 years can be approximated by the equation $P = 621\,900\,(0.907)^t$, where P represents the median sale price t years from now.
The median national rental price of property over the next 10 years can be approximated by the equation $Q = 320\,(1.053)^t$, where Q represents the median rental price t years from now.

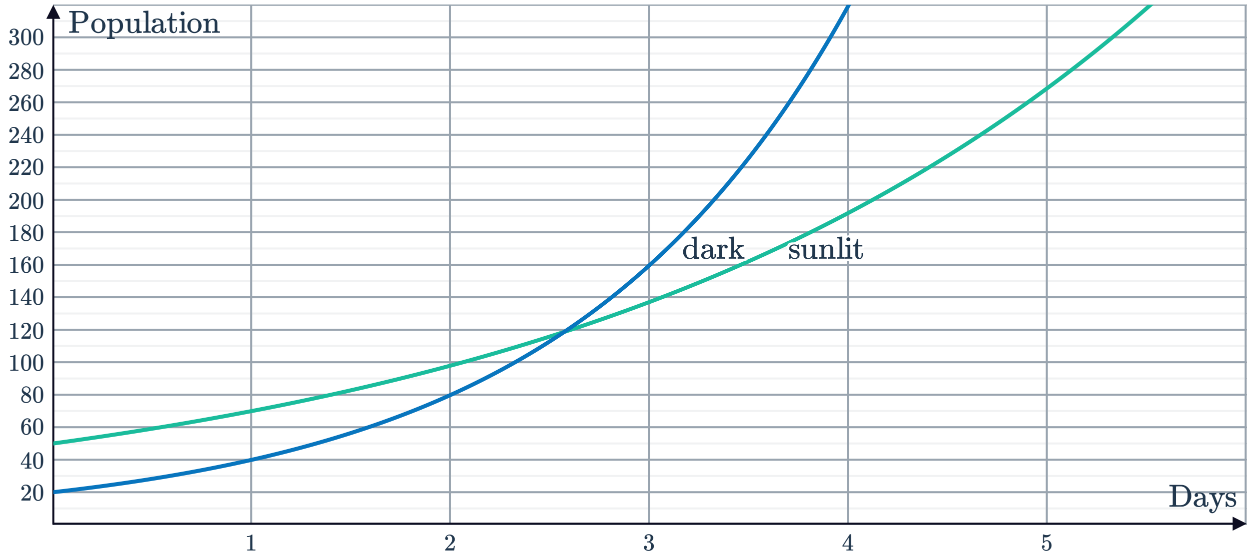
- **a** According to the models, which will decrease over the next 10 years?
- **b** At what annual percentage rate are rental prices expected to increase over the next 10 years?
- **c** If interest rates decrease at any time, the annual percentage change in sale prices is expected to increase by 4%. If this occurs, what will be the new annual percentage change in sale prices?

7 Mohamad has just started a new job, where his starting salary is $\$60\,000$ p.a. and is expected to increase by 3.2% each year. Elizabeth has also just started a new job, which has a starting salary of $\$50\,000$ p.a. and is expected to increase at a rate of 6.1% each year.

- **a** Write an equation for M, Mohamad's salary in t years time.
- **b** Write an equation for P, Elizabeth's salary in t years time.
- **c** Calculate who has the greater salary in 6 years time, and by how much.



8 To investigate the environmental effect on bacterial growth, two colonies of the same bacteria were placed one in a constantly sunlit environment, the other in a dark environment. The graph shows the population of each colony after a certain number of days.



a Identify how many more bacteria were initially present in the sunlit environment compared to the dark environment.

b Determine the percentage rate by which the bacteria left in sunlight increase each day.

c Determine the percentage rate by which the bacteria left in darkness increase each day.

d Initially there were more bacteria in the sunlit environment. Compare the long term behavior of both colonies.

9 Kenneth currently works for a company where he sees 90 clients and the number of clients increases by approximately 5 each month. He is considering opening his own business which he knows his 90 clients will follow him to. He anticipates that the number of clients at his own business will grow by 4% each month.

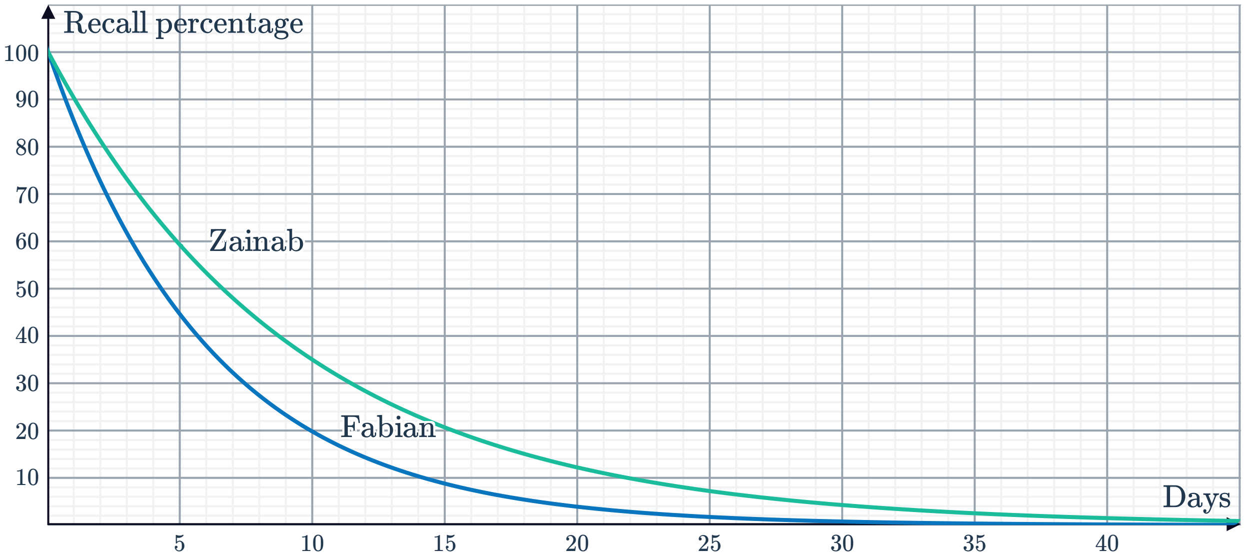
a Form an equation for P, the number of clients he will have in t months if he stays at the company.

b Form an equation for Q, the number of clients he will have in t months if he opens his own business.

c Determine whether Kenneth will have more clients one year from now at his current company or his own business.

d In the long term, determine the option that will allow Kenneth to increase his number of clients at an increasing rate.

10 In a memory study, subjects are asked to m[redacted]rize some content and recall as much as they can remember each day thereafter. Each day, Fabian finds that he has forgotten 15% of what he could recount the day before. Zainab also took part in the study. The graph represents the percentage of content that Fabian (blue) and Zainab (green) could remember after t days.



a Form a model for the percentage P that Fabian can still recall after t days.

b Identify who was forgetting more rapidly.

c Determine if either of them will completely forget the content. Explain your answer.

11 The population of a country is currently estimated at $2\,410\,000$ and is expected to grow by 2.2% each year. The country's food supply is currently enough for a population of $3\,374\,000$ people and is set to increase by 2% each year.

a Determine if there will ever be a food shortage. Explain your answer.

b Explain how you could use a graph to determine when a shortage may occur.

12 Amon and Blair are taking part in a study on blood sugar levels and are asked to eat the same food and in equal quantity, with the only difference being that Amon incorporates lots of sugar into his diet while Blair consumes limited amounts of sugar in their diet.
After eating, their blood sugar level (amount of sugar in the bloodstream) peaked, and was then continually measured, with the following functions modeling their blood sugar over time:
Amon: $f\left(t\right)=179\left(0.3\right)^{t}$, where $f\left(t\right)$ represents blood sugar level at time t hours.
Blair: $g\left(t\right)=132\left(0.6\right)^{t}$, where $g\left(t\right)$ represents blood sugar level at time t hours.

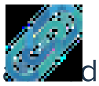
a Determine the value that Amon's blood sugar level peaked.

b Determine the value that Blair's blood sugar level peaked.

c Identify which person's blood sugar level decreased at a slower rate.

d Determine the percentage by which Amon's blood sugar level decreased each hour.

e Explain how the amount of sugar consumed effects your blood sugar levels over time, according to these models.

13 The population of two different bacteria, lad J and K, are given by the following table of values:

Bacteria J:

t (time in days)	0	1	2	3	4
P (population)	1	60	3600	2.16×10^5	1.296×10^7

Bacteria K:

t (time in days)	0	1	2	3	4
Q (population)	1	50	2500	1.25×10^5	6.25×10^6

a Determine which population of bacteria increases faster.

b The population P of bacteria J at time t can be modeled using the general equation $P(t) = a^t$, where $a > 1$. By using the table of values, graph $P(t)$ for $t > 0$.

c The population Q of bacteria K at time t can be modeled using the general equation $Q(t) = b^t$, where $b > 1$. By using the table of values, graph $Q(t)$ for $t > 0$.